

S. S. Rainbow Cake Outbreak Report

Your name here

2026-01-21

Introduction

D.N.P. Cruise Lines has recently emerged as a premier, budget-friendly cruise line that operates out of the port of Halifax, Nova Scotia. Shortly after departing Halifax on February 10th, passengers started reporting with symptoms of illness.

An epidemiological investigation has been on-going with the assistance of the ship's crew. The following report outlines the current status of the investigation and any leads that have been identified to date.

Case Summary

Please draft a short summary of cases in this section that includes a breakdown of the following:

1. Total number of cases who have reported to the ship infirmary
2. Onset date range of cases
3. Case breakdown by sex
4. Case breakdown by age range

Case Symptoms and case definition

Please describe symptoms being experienced by cases, and provide the case definition for both cases reporting **general illness** and those reporting more **serious illness**.

Provide a short statement about the concerns surrounding the **serious illness** cases and some examples of infectious disease that may cause them.

Please provide total number of cases who meet each of the two case definitions either in tabular or text format.

Epidemiologic curve

Please embed your epidemiological curve of serious illnesses from Module 3 here, by including the code in the r-chunk below. We have included an extract of the `symptoms_manifest` dataset in the `participant_data/module4` folder for you to use.

(Note: Normally, we would make several edits to the curve to have it look more like a proper epi-curve. For the purposes of this exercise though, replicating the code exactly how we did it in Module 3 is sufficient. If you'd like to challenge your self, however, please feel free to modify the curve how you see fit.)

Provide a brief interpretation of the epidemiologic curve. In your interpretation, include why the curve appears to indicate a point-source outbreak, and provide additional details from your summaries including the **boarding dates** and **membership status** of passengers experiencing **serious symptom outcomes**.

Food investigation summary

Provide a brief description about the information you have received regarding the meal services provided on February 9th.

Include the table of attack rates for general and serious illness outcomes for each menu item.

Indicate which menu items appear to be the most risky for causing illnesses aboard the ship.

Recommendations and next steps

Using your best epidemiological skills, provide any recommendations you can to the captain of the ship, and outline any next-steps that you would undertake at this point in the cruise.

Indicate any additional questions or concerns that you have about the epidemiologic investigation so far, and highlight any missing information.